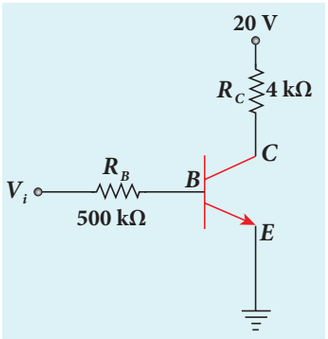



$$I_B = \frac{V_i}{R_B} = \frac{20\,V}{500\,k\Omega} = 40\,\mu A \qquad [\because V_{BE} = 0V]$$

$$I_C = \frac{V_{CC}}{R_C} = \frac{20\,V}{4\,k\Omega} = 5\ mA \qquad [\because V_{CE} = 0V]$$

$$\beta = \frac{I_C}{I_B} = \frac{5\,mA}{40\,\mu A} = 125$$

## 10.4.5 Operating Point

The operating point is a point where the transistor can be operated efficiently. A straight line drawn by joining the points $A\,(V_{CC}, 0)$ and $B\,(0, V_{CC}/R_C)$ is called the DC load line. The DC load line superimposed on the output characteristics of a transistor is used to learn the concept of operating point of the transistor as shown in Figure 10.34.

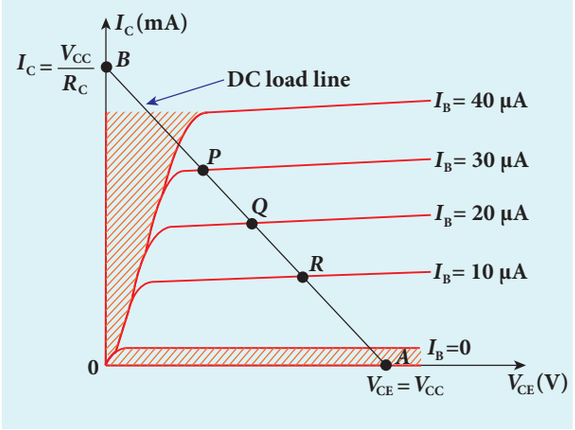



**Figure 10.34** Output characteristics of a transistor in common emitter mode with the DC load line

In Figure 10.34, the points P, Q, R are called Q points or quiescent points which determine the operating point or the working point of a transistor. If the operating point is chosen at the middle of the DC load line (point *Q*), the transistor can effectively work as an amplifier. The operating point determines the maximum signal that can be obtained without being distorted.

For a transistor to work as a open switch, the *Q* point can be chosen at the cut-off region and to work as a closed switch, the *Q* point can be chosen in the saturation region.

## 10.4.6 Transistor as a switch

A transistor in saturation region acts as a closed switch while in cut-off region; it acts as an open switch. It functions like an electronic switch that helps to turn ON or OFF a given circuit by a small control signal which keeps the transistor either in saturation region or in cut-off region. The circuit is shown in Figure 10.35.

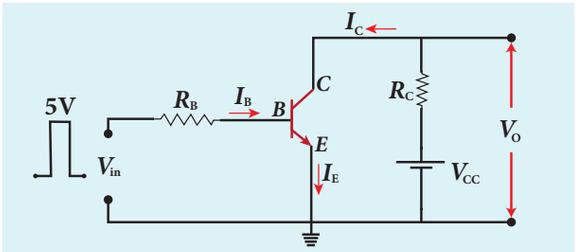



**Figure 10.35** Transistor as a switch

- **When the input is low:**

When the input is low (say 0V), the base current is zero and transistor is not properly forward biased. It is in cut off region. As a result, the collector current is zero and correspondingly the voltage drop across $R_C$ also becomes nearly zero. The output voltage is high and is equal to $V_{CC}$. It means that the no current flows through the transistor and it is said to be switched off. The transistor acts as an open switch.



XII_Physics_Vol-2_UNIT-10_Part 1_EM.indd 222 19-12-2022 16:39:08